Operating Systems

Unit-1

Introduction to Operating Systems

Contents

Prerequisites

- Computer System and Architecture
- Programming in C
- Data Structures and Algorithms

What is Operating System?

Where does the Operating System Sit?

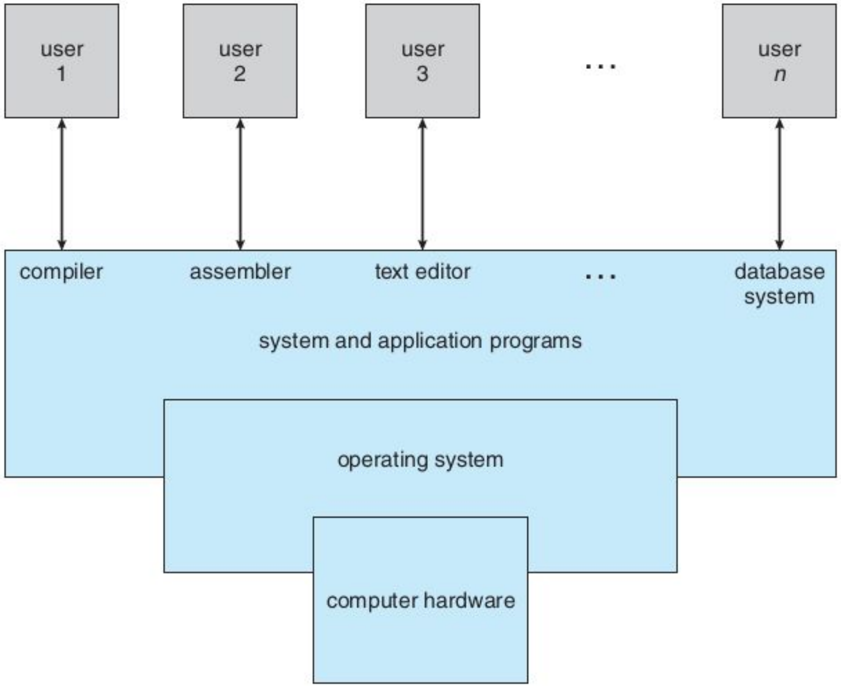


Figure 1.1 Abstract view of the components of a computer system.

Need of Operating System?

- Can we not interact to the Hardware without an OS?
- If we can, why do we need an OS?

The Reason why we need an Operating System?

When There is No OS:

- the user will have to write a program every time he/she wants to access the hardware resources and perform any operation using it.
- Multiple users accessing the resources can not be handled in an efficient way.

Operating System

- It is a System Software which provides an interface between User and the Hardware by hiding th e complexities following the hardware and interactions with it using machine codes.

- It is also the software which provides a platform/environment for other applications and softwares to run.

- It is the entity that not only provides security between users but also controls the access from programs

- As a whole, Operating System is the set of programs that govern the computer system.

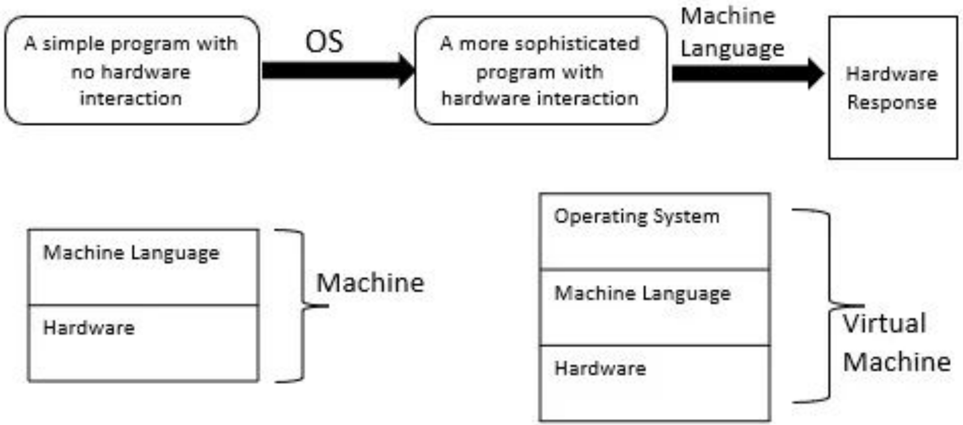
So, what are the main Objectives of Operating Systems?

1. Operating System as an Extended Machine

2. Operating System as an Resource Manager

So, what are the main Objectives of Operating Systems?

1. Operating System as an Extended Machine

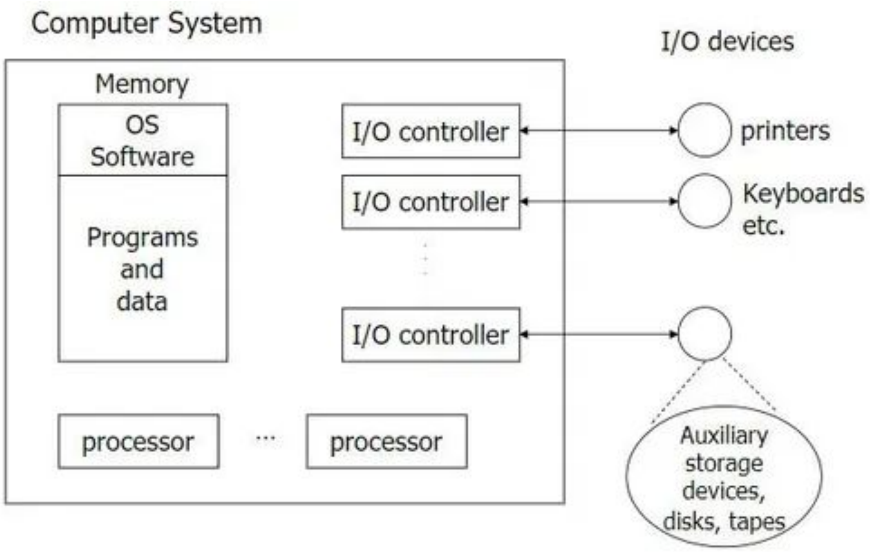


Key Points:

1. Simplification of Hardware Complexity
2. Abstraction of layers
3. Driver development
4. Improved Usability via UIs

So, what are the main Objectives of Operating Systems?

2. Operating System as an Resource Manager



Key Points:

1. Resource Allocation
2. Resource Sharing
3. Conflict Management
4. Efficiency

Examples of Resource Management:

1. CPU Scheduling
2. Memory Management
3. File System Management
4. I/O Device Management

Let's Look into the History and Evolution of OS

1. First Generation (1945-55)

Key Developments:

- **First Digital Computers:** The first functioning digital computer, built by Professor John Atanasoff and Clifford Berry at Iowa State University, used 300 vacuum tubes.
- **Z3 Computer:** Konrad Zuse in Berlin built the Z3 computer using electromechanical relays.
- **Colossus (1944):** Built at Bletchley Park, England, and programmed by a group of scientists including Alan Turing.
- **Mark I (1944):** Built by Howard Aiken at Harvard University.
- **ENIAC (1945):** Constructed by William Mauchley and J. Presper Eckert at the University of Pennsylvania.

1. First Generation (1945-55)

Characteristics:

- **Primitive Technology:** Used vacuum tubes, which were unreliable and required frequent maintenance.
- **Manual Operation:** Programming was done in absolute machine language or by wiring electrical circuits using plugboards.
- **No Operating Systems:** Early computers did not have operating systems. Each machine was manually operated by programmers who signed up for time slots.
- **Punched Cards:** By the early 1950s, punched cards were introduced, allowing programs to be written on cards and read into the computer.

1. First Generation (1945-55)

Positives:

- **Introduction to Digital Computing:** Laid the foundation for digital computing.
- **Programmability:** Some early machines were programmable, albeit in a very primitive manner.

Negatives:

- **Technological Limitations:** Vacuum tube technology hindered performance and reliability.
- **Inefficient Programming:** Manual programming methods were time-consuming and error-prone.
- **Lack of User Friendliness:** Computers were difficult to operate and maintain.

2. Second Generation (1955-65)

Key Developments:

- **Transistors:** Replaced vacuum tubes, resulting in more reliable, smaller, and energy-efficient computers.
- **Mainframes:** Large, expensive machines housed in specially air-conditioned rooms, operated by professional staff.
- **Separation of Roles:** Clear distinction between designers, builders, operators, programmers, and maintenance personnel.
- **Batch Processing Systems:** Introduced to minimize wasted computer time and optimize resource utilization.

2. Second Generation (1955-65)

Characteristics:

- **Mainframes:** Only large organizations could afford these multimillion-dollar machines.
- **Programming with Punched Cards:** Programmers wrote programs on paper (in FORTRAN or assembly language), punched them onto cards, and handed the card decks to operators.
- **Batch Processing:** Jobs were collected in batches and processed sequentially to reduce idle time.

2. Second Generation (1955-65)

Example:

- **GM-NAA I/O**: Considered one of the first operating systems, developed for the IBM 704
- **FMS (Fortran Monitor System):** A batch processing system designed to handle FORTRAN programs.
- **IBSYS:** IBM's operating system for the IBM 7094, which supported batch processing.
- **Atlas Supervisor**: Developed for the Atlas Computer, one of the first computers with virtual memory.

Applications:

- **Scientific and Engineering Calculations:** Used for solving partial differential equations in physics and engineering.
- **Programming Languages:** FORTRAN and assembly language were predominantly used.

2. Second Generation (1955-65)

Positives:

- **Increased Reliability:** Transistors significantly improved the reliability of computers compared to vacuum tubes.
- **Resource Optimization:** Batch processing systems reduced idle time and optimized the utilization of expensive computer resources.

Negatives:

- **Limited Interaction:** Batch processing limited user interaction with the computer during job execution.
- **Complex Job Submission:** Submitting jobs required physical handling of punched cards and magnetic tapes, which was time-consuming and prone to errors.

3. Third Generation (1956-80)

Key Developments:

- **Integrated Circuits (ICs):** These miniaturized circuits replaced bulky transistors, resulting in smaller, faster, and more affordable computers.
- **IBM System/360:** A successful family of computers with varying processing capabilities, promoting software compatibility across different models.
- **Multiprogramming:** This technique allows running multiple programs concurrently, maximizing CPU utilization by executing one program while another waits for I/O operations.
- **Spooling (Simultaneous Peripheral Operation On Line):** This approach enables continuous job processing by reading incoming jobs onto disk and spooling output to a different device, eliminating the need for dedicated tape handling.

3. Third Generation (1956-80)

Characteristics

- Compatibility across a series of machines (System/360) from small to large scales.
- Multiprogramming: memory partitioning to hold multiple jobs simultaneously.
- Spooling: reading jobs onto disks for efficient job management.
- Timesharing: providing fast, interactive service to multiple users through online terminals.
- Enormous and complex operating systems (e.g., OS/360) supporting diverse requirements and environments.

3. Third Generation (1956-80)

Examples

- **IBM System/360**: Software-compatible machines ranging from small to large.
- **OS/360**: Operating system designed to work on all models of the System/360.
- **CTSS (Compatible Time Sharing System)**: Developed at M.I.T. for interactive use.
- **MULTICS (MULTiplexed Information and Computing Service)**: Aimed to support hundreds of users simultaneously.

Applications

- **Commercial Data Processing**: Managing huge databases, tape sorting, printing, airline reservations.
- **Scientific Calculations**: Weather forecasting, engineering computations.
- **Timesharing**: Interactive use by multiple users for program development and debugging.
- **Batch Systems**: Large-scale data processing tasks in commercial and scientific domains.

3. Third Generation (1956-80)

Positives:

- **Compatibility**: Programs written for one machine in the series could run on others.
- **Multiprogramming**: Increased CPU utilization by running multiple jobs simultaneously.
- **Spooling**: Efficient job management and elimination of intermediate tape handling.
- **Timesharing**: Fast, interactive service for multiple users.
- **Cost-Effectiveness**: Use of ICs reduced the cost and increased the performance of computers.

Negatives:

- **Complexity**: Enormous and complex operating systems with thousands of bugs.
- **Resource Management**: Difficulty in writing software that efficiently met diverse and conflicting requirements.
- **Debugging Delays**: Batch processing caused long delays between job submission and output, leading to inefficient debugging.
- **Commercial Failure**: Ambitious projects like MULTICS faced commercial challenges despite technical advancements.

4. Fourth Generation (1980-present)

Key Developments:

- **Microprocessors**: The development of microprocessors, such as the Intel 8080, laid the foundation for personal computing.
- **Personal Computers**: The introduction of PCs, exemplified by the Apple II and IBM PC, brought computing power to individuals.
- **Operating Systems**: The development of operating systems like CP/M, MS-DOS, and Windows catered to the needs of personal computer users.
- **Graphical User Interfaces (GUIs)**: Pioneered by Xerox PARC and popularized by Apple, GUIs made computers more user-friendly.
- **Networking**: The emergence of networks, such as the internet, facilitated communication and resource sharing among computers.

4. Fourth Generation (1980-present)

Characteristics:

- **Microprocessors**: Integrated circuits containing thousands of transistors on a single chip, significantly reducing the size and cost of computers.
- **Personalization**: Computers became more affordable and accessible, leading to widespread use in homes and businesses.
- **User-Friendly Interfaces**: GUIs allowed users to interact with computers through graphical icons and visual indicators rather than text commands.
- **Networking**: Enhanced connectivity through local and wide-area networks, and the growth of the internet.
- **Software Development**: Rapid growth in software applications, including productivity tools, games, and business applications.

4. Fourth Generation (1980-present)

Examples

- **CP/M (Control Program for Microcomputers)**: An early operating system for microcomputers, widely used before the rise of MS-DOS.
- **MS-DOS (Microsoft Disk Operating System)**: Became the standard OS for IBM PCs.
- **Apple DOS**: Used on the Apple II series of computers.
- **Windows**: Microsoft's graphical operating system, starting with Windows 1.0 and evolving into the dominant desktop OS.
- **Mac OS**: Apple's operating system for Macintosh computers, starting with System 1 and evolving into macOS.

4. Fourth Generation (1955-65)

Applications

- **Home Computing**: PCs for personal use, including education, entertainment, and household management.
- **Business and Office**: Productivity software like word processors, spreadsheets, and databases revolutionized business operations.
- **Networking and Communication**: Email, instant messaging, and web browsing became integral to personal and professional communication.
- **Education**: Computers in schools and universities for research, teaching, and learning.
- **Entertainment**: Video games, multimedia, and digital content creation became popular uses for personal computers.

4. Fourth Generation (1955-65)

Positives:

- **Affordability**: The reduced cost of microprocessors made computers accessible to the general public.
- **Ease of Use**: GUIs made computers easier to use for people without technical expertise.
- **Connectivity**: Networking capabilities allowed for greater communication and resource sharing.
- **Flexibility**: Wide range of software applications catered to various user needs and preferences.
- **Innovation**: Rapid technological advancements spurred continuous improvement and innovation in computing.

Negatives:

- **Software Compatibility**: Issues with running software across different platforms and operating systems.
- **Security and Privacy**: Increased risk of cyber threats and challenges in protecting personal information.
- **Obsolescence**: Rapid technological advancements led to quick obsolescence of hardware and software.
- **Learning Curve**: Despite GUIs, some users still faced challenges in adapting to new technologies.
- **Maintenance**: Personal computers required regular maintenance and updates to ensure optimal performance.

5. Fifth Generation (1990-present)

Key Developments

- **Mobile Phones**: The evolution from bulky devices to smartphones with internet connectivity and computing capabilities.
- **Smartphones**: The convergence of mobile phones and personal digital assistants (PDAs) into powerful handheld computers.
- **Mobile Operating Systems**: The development of operating systems specifically designed for mobile devices, such as iOS, Android, and Windows Phone.
- **App Stores**: Platforms for distributing mobile applications, creating a thriving ecosystem of developers and users.
- **Cloud Computing**: Integration of mobile devices with cloud services for storage, computation, and synchronization.

5. Fifth Generation (1990-present)

Characteristics

- **Portability**: Mobile devices are compact and portable, allowing users to carry computing power wherever they go.
- **Touch Interfaces**: Touchscreen technology has become the primary mode of interaction with mobile devices.
- **App Ecosystem**: A vast ecosystem of mobile applications provides diverse functionalities, from productivity to entertainment.
- **Constant Connectivity**: Mobile devices are always connected through cellular networks and Wi-Fi, enabling real-time communication and access to information.
- **Integrated Sensors**: Devices come with built-in sensors like GPS, accelerometers, and cameras, enhancing their capabilities.

5. Fifth Generation (1990-present)

Examples

- **iOS**: Apple's operating system for iPhone and iPad, known for its App Store and user-friendly interface.
- **Android**: Google's open-source operating system, the most widely used mobile OS worldwide.
- **Windows Phone**: Microsoft's mobile operating system, aimed at integrating with Windows desktop environments.
- **BlackBerry OS**: An operating system for BlackBerry smartphones, known for its secure email services.
- **Tizen**: An open-source operating system primarily used by Samsung in various devices, including smartwatches and smart TVs.

5. Fifth Generation (1990-present)

Applications:

- **Communication**: Instant messaging, video calls, and social media apps facilitate global communication.
- **Productivity**: Mobile apps for email, document editing, and project management support work on the go.
- **Entertainment**: Streaming services, mobile games, and multimedia apps provide endless entertainment options.
- **Navigation**: GPS-enabled apps assist with directions, location-based services, and travel planning.
- **Health and Fitness**: Apps for tracking physical activity, monitoring health metrics, and providing medical information.

5. Fifth Generation (1990-present)

Positives:

- **Convenience:** Mobile devices offer unprecedented convenience and accessibility, fitting powerful computing capabilities into a pocket-sized form.
- **Real-Time Access**: Constant connectivity allows for real-time access to information, communication, and services.
- **Versatility**: A wide range of applications caters to diverse needs, from personal use to professional tasks.
- **Innovation**: The competitive mobile market drives rapid innovation in hardware and software.
- **Integration**: Seamless integration with cloud services enhances storage, synchronization, and computational capabilities.

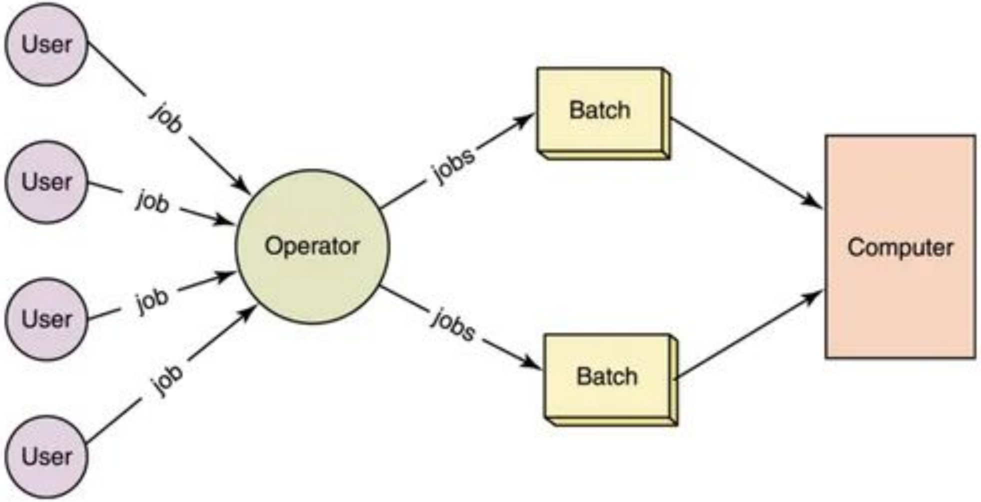
Negatives:

- **Battery life:**HIgh usage and constant connectivity can drain battery life quickly, limiting device availability.
- **Security and Privacy**: Mobile devices are susceptible to security threats, data breaches, and privacy concerns.
- **Fragmentation**: Different operating systems and device specifications can lead to compatibility issues for apps.
- **Distraction**: The constant presence of mobile devices can lead to distractions and reduced productivity.
- **Obsolescence**: Rapid technological advancements result in frequent obsolescence of devices and software.

Types of OS

1. Batch Operating System
2. Multiprogramming Operating System
3. Multiprocessor Operating System
4. Time-Sharing Operating System
5. Real-Time Operating System
6. Distributed Operating System
7. Network Operating System
8. Embedded Operating System

Batch Operating System



Batch Operating System

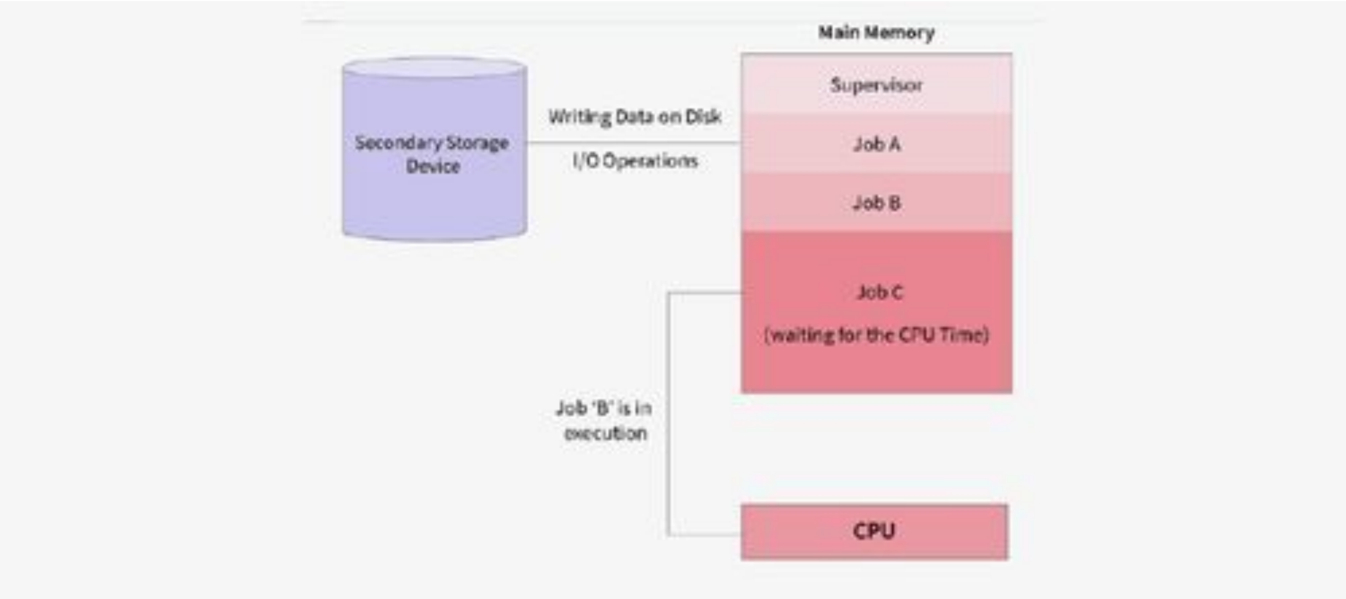
Advantages of Batch Operating System

1. **Efficiency:** Batch systems can be very efficient when jobs require similar resources, as they can be executed without interruption.
2. **Reduced Setup Time:** Jobs are collected and processed in batches, reducing the need for frequent manual intervention.
3. **Resource Utilization:** The system can optimize resource usage by grouping jobs with similar requirements.
4. **Automation:** Once a batch is submitted, the system handles the execution without further user intervention.

Disadvantages of Batch Operating System

1. **No User Interaction:** Users cannot interact with their jobs once they are submitted, making it difficult to correct errors during execution.
2. **Debugging Difficulty:** Diagnosing and fixing errors can be challenging since there is no interaction during job execution.
3. **Job Prioritization:** Jobs are typically executed in the order they are received, which might not prioritize urgent tasks.
4. **Idle Time:** There can be significant idle time if the system is waiting for a batch to complete before starting the next.

Multiprogramming Operating System



Multiprogramming Operating System

Advantages of Multiprogramming Operating System

1. **Increased CPU Utilization:** Ensures the CPU is always busy by switching to another job when one job is waiting for I/O operations.
2. **Reduced Idle Time:** Minimizes idle time by overlapping I/O and CPU activities.
3. **Improved Throughput:** Increases the number of jobs completed in a given time frame.
4. **Better Resource Utilization:** Optimizes the use of system resources (CPU, memory, I/O devices).

Disadvantages of Multiprogramming Operating System

1. **Complexity:** Increased complexity in managing multiple jobs and resources.
2. **Memory Management:** Requires efficient memory management to handle multiple jobs in memory.
3. **Context Switching Overhead:** Frequent context switching can introduce overhead, reducing efficiency.
4. **Resource Contention:** Potential for resource contention when multiple jobs compete for the same resources.

Multiprocessor Operating System

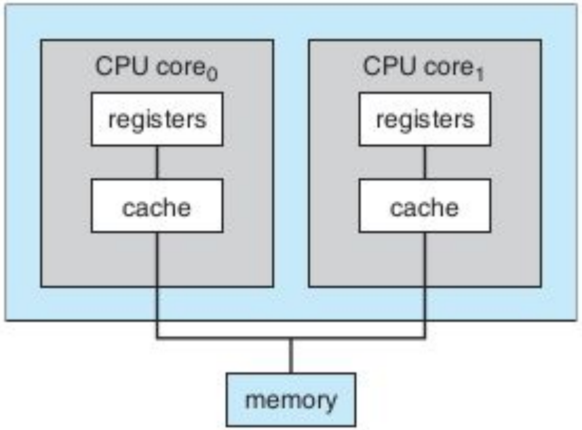


Figure 1.7 A dual-core design with two cores placed on the same chip.

Multiprocessor Operating System

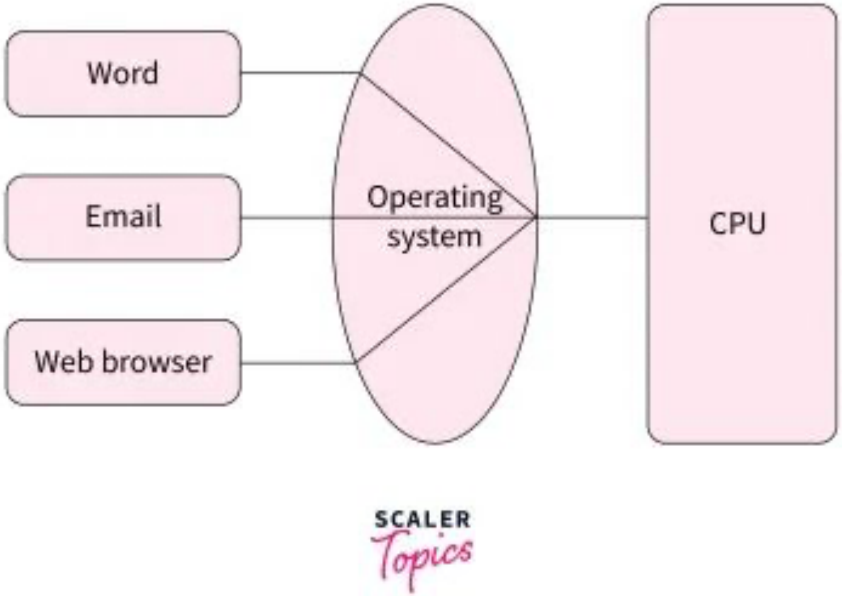
Advantages of Multiprocessor Operating System

1. **Increased Throughput:** Multiple processors allow the system to handle more tasks simultaneously, increasing overall throughput.
2. **Improved Performance:** Parallel execution of processes reduces execution time, improving system performance.
3. **Scalability:** Additional processors can be added to the system to handle increased workloads.
4. **Fault Tolerance:** If one processor fails, others can continue to work, improving system reliability and availability.

Disadvantages of Multiprocessor Operating System

1. **Complexity:** Managing multiple processors and ensuring efficient load balancing adds complexity to the operating system.
2. **Cost:** Multiprocessor systems are more expensive due to the additional hardware.
3. **Resource Contention:** Multiple processors accessing shared resources can lead to contention and potential performance bottlenecks.
4. **Synchronization Issues:** Coordinating tasks between processors requires complex synchronization mechanisms to ensure data consistency and prevent race conditions.

Time-Sharing Operating System



Time Sharing Operating System

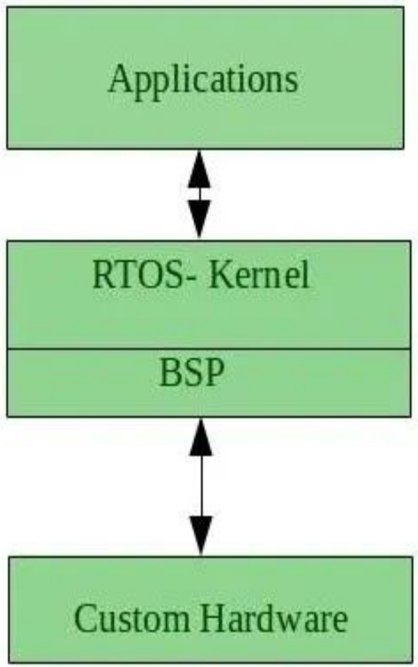
Advantages of Time-Sharing Operating System

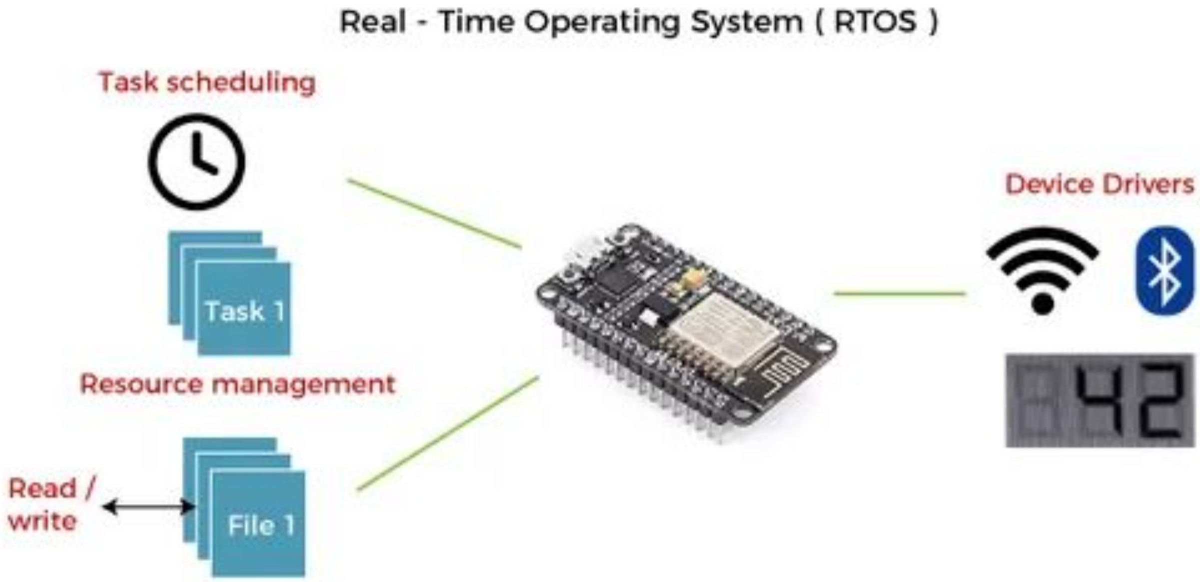
1. **Interactive Computing:** Provides a responsive environment where users can interact with their applications in real time.
2. **Efficient Resource Utilization:** Maximizes CPU utilization by sharing CPU time among multiple users and processes.
3. **Fairness:** Ensures that all users get a fair share of CPU time.
4. **Improved Productivity:** Allows multiple users to work on the system simultaneously, improving overall productivity.

Disadvantages of Time-Sharing Operating System

1. **Overhead:** Context switching introduces overhead, reducing overall system efficiency.
2. **Complexity:** Managing multiple users and processes requires sophisticated scheduling and resource management.
3. **Security and Isolation:** Ensuring security and isolation between users' processes is challenging.
4. **Resource Contention:** Multiple users sharing the same resources can lead to contention and potential performance degradation.

Real-Time Operating System





Real Time Operating System

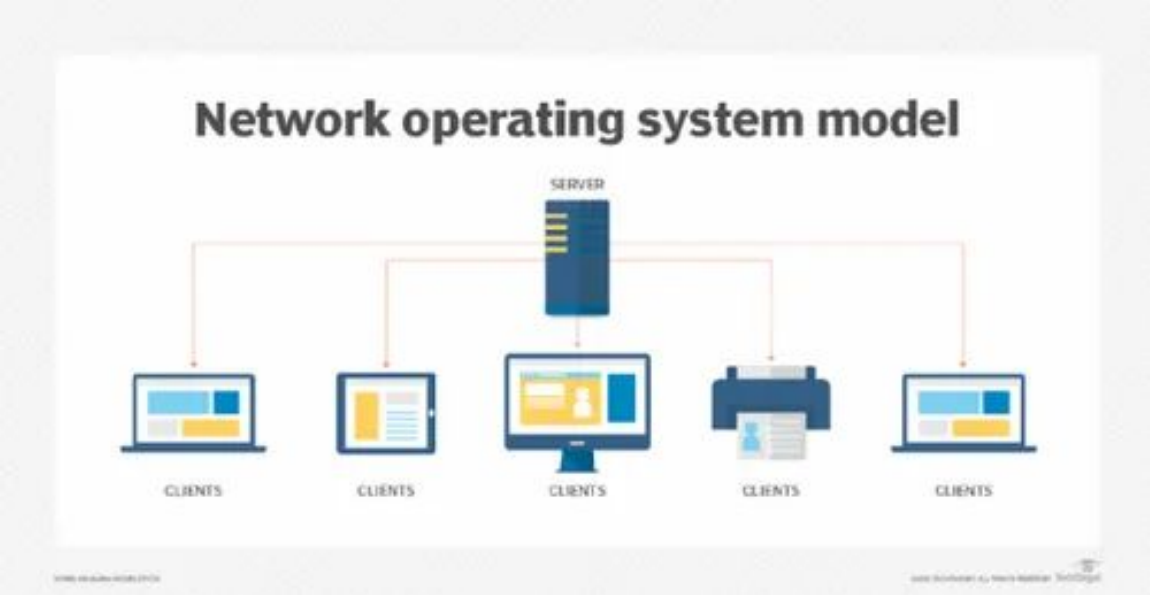
Advantages of Real-Time Operating System

1. **Predictability:** Ensures predictable behavior and timely task execution.
2. **High Reliability:** Suitable for critical applications requiring high reliability and precise timing.
3. **Low Latency:** Minimal interrupt latency and context switching time ensure quick response to events.
4. **Priority Management:** Efficient handling of high-priority tasks ensures critical tasks are not delayed.

Disadvantages of Real-Time Operating System

1. **Complexity:** Designing and implementing an RTOS is complex due to the need for precise timing and scheduling.
2. **Cost:** Higher development and maintenance costs compared to general-purpose operating systems.
3. **Limited Flexibility:** RTOS is typically specialized for specific applications, reducing its flexibility for other tasks.
4. **Resource Constraints:** Often runs on systems with limited resources, requiring efficient use of CPU, memory, and power.

Network Operating System



Network Operating System

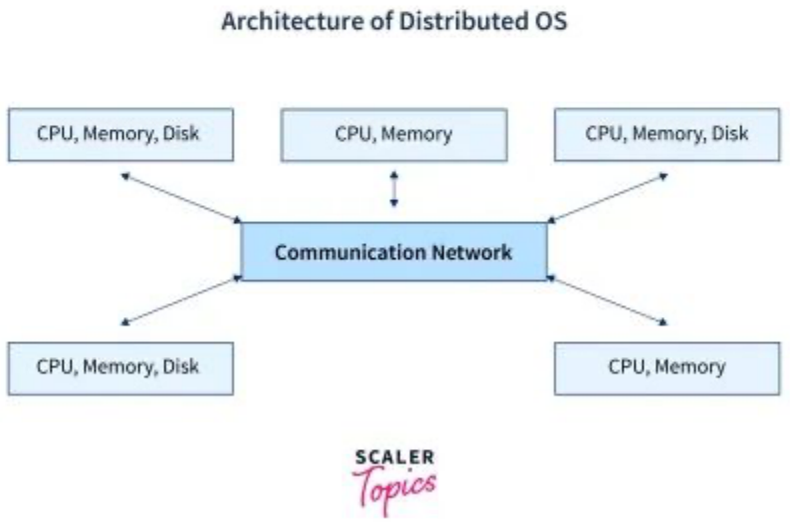
Advantages of Network Operating System

1. **Centralized Management:** Simplifies administration and management of network resources.
2. **Resource Sharing:** Efficiently shares resources like files, printers, and applications across the network.
3. **Improved Security:** Provides robust security measures, including user authentication and access controls.
4. **Reliability:** Enhances network reliability and uptime through centralized control and management.
5. **Scalability:** Easily scales by adding more servers and clients to the network.

Disadvantages of Network Operating System

1. **Cost:** Higher initial setup and maintenance costs due to the need for servers and network infrastructure.
2. **Complexity:** Requires skilled network administrators to manage and maintain the network.
3. **Dependency:** Network performance and availability depend on the server and network infrastructure.
4. **Single Point of Failure:** Centralized servers can become single points of failure, impacting network availability.

Distributed Operating System



Distributed Operating System

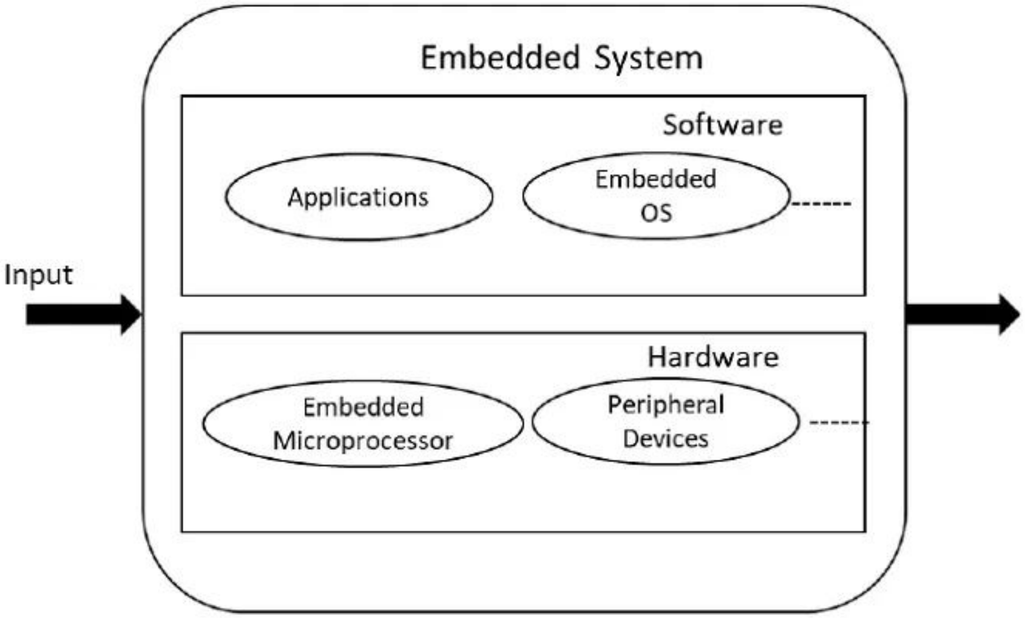
Advantages of Distributed Operating System

1. **Resource Sharing:** Efficient sharing of resources across the network, improving utilization and performance.
2. **Scalability:** Easily scales by adding more nodes to the network.
3. **Fault Tolerance:** Enhanced reliability and fault tolerance through redundancy and failover mechanisms.
4. **Parallelism:** Enables parallel processing, improving performance for large-scale computations.
5. **Transparency:** Provides a seamless and transparent interface, abstracting the complexities of the underlying hardware and network.

Disadvantages of Distributed Operating System

1. **Complexity:** Increased complexity in managing and coordinating multiple nodes and resources.
2. **Network Dependency:** Performance and reliability are dependent on the underlying network infrastructure.
3. **Security:** Ensuring security across a distributed environment is challenging.
4. **Consistency:** Maintaining data consistency and coherence across multiple nodes can be difficult.
5. **Latency:** Network communication can introduce latency, affecting performance.

Embedded Operating System



Embedded Operating System

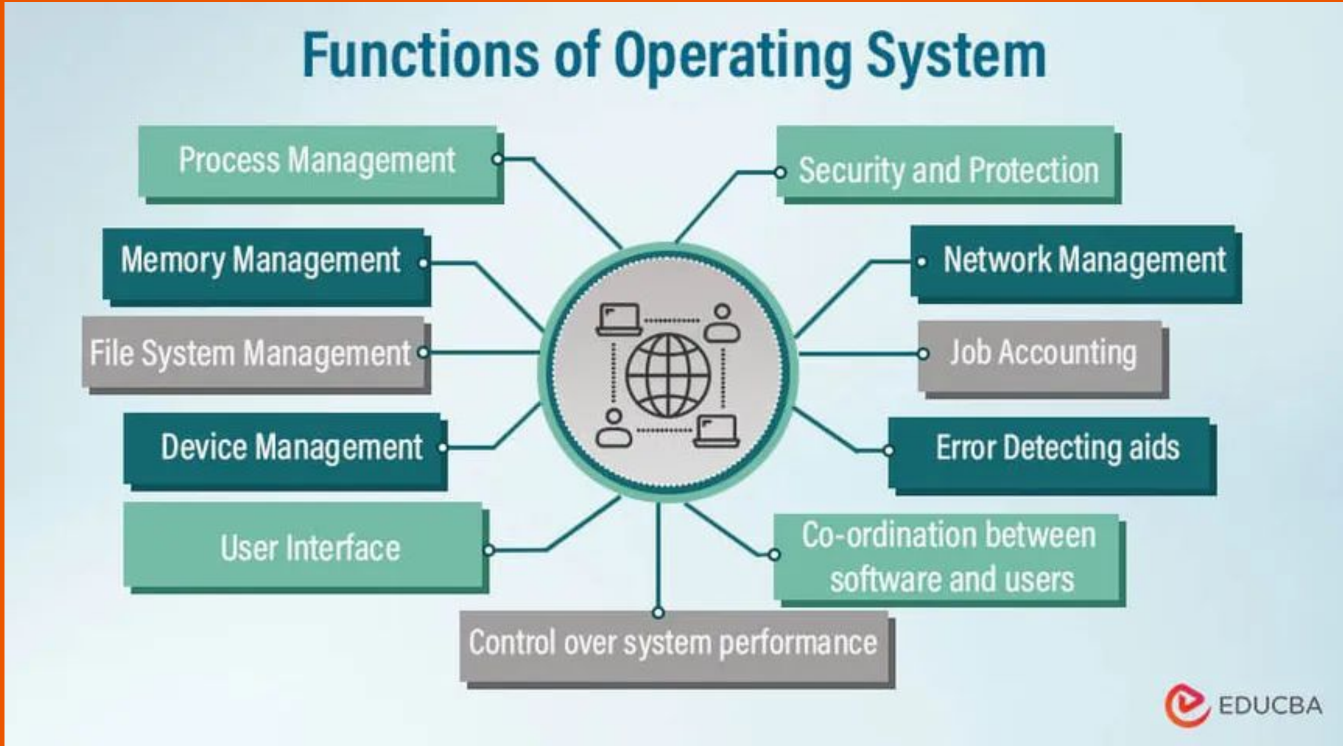
Advantages of Embedded Operating System

1. **Efficiency:** Optimized for performance and resource usage, ensuring efficient operation within constrained environments.
2. **Real-Time Processing:** Provides timely and predictable responses, essential for applications requiring precise timing and control.
3. **Reliability:** Designed to be highly reliable and stable, crucial for embedded applications that require continuous operation.
4. **Specialization:** Tailored to the specific needs of the embedded application, offering specialized functionalities and interfaces.

Disadvantages of Embedded Operating System

1. **Limited Functionality:** May lack the extensive features and flexibility of general-purpose operating systems.
2. **Complex Development:** Developing and maintaining an EOS can be complex, requiring specialized knowledge of the embedded hardware and software.
3. **Resource Constraints:** Limited resources can restrict the scope of applications and functionalities.
4. **Upgrade Difficulty:** Upgrading or modifying an embedded system can be challenging, especially in deployed products.

Function of OS



Functions of OS:

- Process Management,
- Memory Management,
- File System Management,
- Device Management,
- User Interface,
- Control over System Performance,
- Co-ordination between software and user,
- Error Detecting Aids,
- Job Accounting,
- Network Management,
- Security and Protection.

Process Management

Process Scheduling: Allocating CPU time to various processes using scheduling algorithms (e.g., round-robin, priority scheduling).

Process Creation and Termination: Creating and terminating processes, including handling system calls like `fork()` and `exit()`.

Process Synchronization: Ensuring processes do not interfere with each other using synchronization mechanisms like semaphores, mutexes, and monitors.

Inter-Process Communication (IPC): Facilitating communication between processes through pipes, message queues, shared memory, and sockets.

Memory Management

Memory Allocation: Allocating and deallocating memory to processes, managing the available memory.

Virtual Memory: Implementing virtual memory to extend the apparent amount of memory by using disk space.

Paging and Segmentation: Using paging and segmentation to manage memory efficiently and protect processes from each other.

Memory Protection: Ensuring that processes do not access memory allocated to other processes.

File System Management

File Creation and Deletion: Providing mechanisms to create, delete, read, and write files.

Directory Management: Organizing files into directories and managing directory structures.

File Access Control: Implementing access control to protect files from unauthorized access.

Storage Management: Managing the underlying storage devices (e.g., hard drives, SSDs) and ensuring data integrity and security.

Device Management

Device Drivers: Providing drivers to interface with hardware devices, abstracting the details of hardware from the user.

I/O Scheduling: Managing the input and output operations and ensuring efficient use of I/O devices.

Buffering and Caching: Using buffering and caching to improve the efficiency of I/O operations.

Device Communication: Facilitating communication between the OS and hardware devices.

Security and Protection

User Authentication: Implementing mechanisms to verify the identity of users (e.g., login systems, biometrics).

Access Control: Enforcing policies to control access to resources based on user permissions.

Encryption: Using encryption to protect data both at rest and in transit.

Security Monitoring: Detecting and responding to security breaches, ensuring system integrity.

Networking

- **Network Protocols:** Implementing network protocols to facilitate communication between computers (e.g., TCP/IP).
- **Network Services:** Providing services such as file sharing, email, and web hosting.
- **Resource Sharing:** Enabling the sharing of resources (e.g., printers, files) across a network.

User Interface

Command-Line Interface (CLI): Providing a text-based interface for users to interact with the system.

Graphical User Interface (GUI): Offering a visual interface for user interaction, including windows, icons, and menus.

System Utilities: Providing system utilities and tools for managing and configuring the OS.

System Performance Monitoring

Performance Metrics: Collecting data on system performance (e.g., CPU usage, memory usage, disk I/O).

Resource Management: Adjusting resource allocation to optimize performance.

System Logs: Maintaining logs of system activities and events for monitoring and troubleshooting.